

Association of attention-deficit/hyperactivity disorder and atopic eczema modified by sleep disturbance in a large population-based sample

Marcel Romanos,¹ Manfred Gerlach,¹ Andreas Warnke,¹ Jochen Schmitt²

¹Department of Child and Adolescent Psychiatry, Psychosomatics and Psychotherapy, University Clinic of Wuerzburg, Wuerzburg, Germany ²Department of Dermatology, Medical Faculty Carl Gustav Carus, Technische Universität Dresden, Dresden, Germany

Correspondence to

Dr Marcel Romanos, University Clinic of Wuerzburg, Fuechsleinstr 15, 97080 Wuerzburg, Germany; romanos@kjp.uni-wuerzburg.de

Accepted 26 June 2009

Published Online First

19 August 2009

ABSTRACT

Background A cross-sectional study was performed in a population-based German sample (n=13 318 children and adolescents aged 3–17) to replicate the recently reported independent association of atopic eczema (AE) and attention-deficit/hyperactivity disorder (ADHD) and to further investigate the role of environmental and behavioural factors.

Methods The odds ratio (OR) was calculated with allergic comorbidity and various environmental factors considered as confounders and sleeping problems hypothesised to act as a potential effect modifier.

Results The lifetime prevalence of AE and ADHD was 14.7% (male 14.4%; female 14.9%) and 4.9% (male 7.8%; female 2.0%), respectively. There was an association between ADHD and AE (OR 1.54; 95% CI 1.24 to 1.93; p<0.001), which was independent of sociodemographic factors, parental smoking, breastfeeding, number of siblings, perinatal health problems, and atopic comorbidity. Further analyses of a subgroup of 6484 children age 3–11 confirmed the hypothesis that the association between AE and ADHD was modified by sleeping problems (interaction effect AE*sleep problems OR 2.02 95% CI 1.03 to 3.97; p=0.04). There was a strong association between AE and ADHD in children with SP (OR 2.67 95% CI 1.51 to 4.71; p=0.001; n=1112), but not in children without SP (OR 1.24 95% CI 0.83 to 1.84; p=0.30; n=5796).

Conclusions ADHD and AE appear to be strongly and independently associated in children with sleeping problems, but not in children without sleeping problems. A substantial part of diagnoses met for ADHD might be engendered by the presence of AE and concomitant sleeping problems.

Attention-deficit/hyperactivity disorder (ADHD) is a frequent and highly persistent paediatric disorder of complex multifactorial aetiology.¹ ADHD affects more than 5% of all children and causes severe psychosocial maladaptation, impairment of quality of life and economic burden on social services.^{2–3} Comorbidity is increasingly being recognised as an important modifier of phenotypic variance, although recent research focused on psychiatric comorbidity present in up to 87% of the children with ADHD.^{4–5} For several decades there has been an ongoing discussion as to whether ADHD is associated with allergic disorders, although various studies have not resulted in conclusive evidence for the assumption.⁶ However, none of those investigations controlled for the presence of atopic eczema (AE), also known as eczema or atopic dermatitis.^{7–8}

AE is the most prevalent chronic-inflammatory condition in children affecting up to 20% of the children at age 6 in western countries.⁹ The typical itchy lesions may cause substantial psychosocial impairment and are a leading cause of sleep loss in childhood.^{10–12} In the subset of patients with extrinsic AE, the itchy lesions often represent the beginning of the atopic march.¹³ However, the term “atopic” might not be thoroughly appropriate, since only approximately 40% of the children with AE have increased IgE levels, and only approximately every third child with AE develops asthma or allergic rhinitis.^{9–14}

Recently we reported a significant association of ADHD and AE in a large population-based sample.¹⁵ However, phenotypic variance of both ADHD and AE is known to be influenced by environmental factors that might account for the observed association.^{9–16–17} In our previous report, we were unable to control for various factors, thus limiting our finding. Here, in order to replicate our finding and to further explore potential confounding and modifying factors, we analysed data from the German Health Interview and Examination Survey for Children and Adolescents (KIGGS), a nationwide representative survey on the health status of German children and adolescents with a vast amount of information on sociodemographics and health-related topics.¹⁸ Since sleep impairment is highly prevalent in both AE and ADHD,^{10–19–20} and disturbed sleep is suggestive of aggravating both symptoms of AE and ADHD,^{21–22} we previously hypothesised that sleep problems may causally link both disorders.^{8–15} Thus, we in particular tested our previously published hypothesis that disturbance of sleep modifies the association between ADHD and AE.^{8–15}

METHODS

We performed a correlational analysis utilising the public use files of the KIGGS study.¹⁸ KIGGS was conducted from 2003 to 2006 in 167 different locations in Germany. From a total of 28 299 subjects asked to participate, 17 642 aged 0–17 years agreed (66.6%). There were no significant differences between participants and children who refused to participate in terms of sociodemographic characteristics and general health status. The public use files are available upon request from the “Robert Koch Institute” (<http://rki.de/>). The study was performed in accordance with the Declaration of Helsinki protocols. Informed consent was not required because all personal identifiers had been removed from the database prior to its use for research. Data

Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

J Epidemiol Community Health: first published as 10.1136/jech.2009.093534 on 19 August 2009. Downloaded from <http://jech.bmj.com/> on September 22, 2026 at SUNY Stony Brook NERL Consortia.

from 13 318 children and adolescents aged 3–17 were included in this study. Age 0–2 years was the single exclusion criterion since, in the KIGGS study, diagnosis of ADHD is hardly reported before age 3. Data from the KIGGS study used in this investigation are based on self-administered questionnaires and standardised parental interviews conducted by trained interviewers. In case of AE, ADHD, and atopic disorders, parents reported a lifetime diagnosis met on the basis of medical examination (for ADHD: medical or psychological examination). ADHD in Germany is generally diagnosed according to the ICD-10 criteria of “hyperkinetic disorder” that largely correspond with the criteria of the DSM-IV-R definition of ADHD combined subtype.^{23 24} We investigated the relationship of AE, ADHD, allergic rhinitis, asthma and a broad spectrum of behavioural and environmental covariates known or suggested to be associated with ADHD and/or AE, such as family income, social status (low/medium/high; based on parental education, professional qualification, professional status, family net income),²⁵ breastfeeding (ever/never), postnatal health problems including breathing problems, maladaptation, infections, jaundice, low birth weight/premature delivery, in-patient treatment (present/absent), maternal smoking during pregnancy (ever/never), exposure to tobacco smoke at home (ever/never), number of siblings (none/one or more), parental history of atopy (absent/at least in one parent) and the living environment (urban vs rural area/small town; East vs West Germany). Logistic regression models (adjusted multivariate analysis) were fitted, testing nested models by using the likelihood ratio test.²⁶ Subjects with missing data in any variable included in the final model were excluded from this analysis. AE was modelled as a binary variable (ever vs never diagnosed) in the analysis. Furthermore, a preplanned hypothesis-driven subgroup analysis on effect modification by the current presence of sleeping problems was performed in a subgroup of $n=6484$ children age 3–11 years with further data available with regard to presence of sleep disturbance based on parental reports. A subsequent stratified analysis was performed for significant interaction results.²⁶ No additional sensitivity analyses were performed. Significance

level was defined at <0.05 . Data were analysed using STATA version 8.

RESULTS

Table 1 summarises the sociodemographic characteristics, data on the lifetime prevalence of ADHD, AE, allergic asthma, allergic rhinitis, family history for atopy, as well as relevant behavioural and environmental characteristics of the study sample. The mean age of the study participants was 10 years. Fifty per cent of the sample was female. The lifetime prevalence of AE and ADHD was 14.7% ($n=1952$; male 14.4%; female 14.9%) and 4.9% ($n=653$; male 7.8%; female 2.0%), respectively. Socio-economic status was higher in children with AE compared with children without AE and lower in children with ADHD compared with children without ADHD (table 1). As expected, AE was significantly related to allergic asthma ($p<0.001$) and allergic rhinitis ($p<0.001$). Children with ADHD also had an increased lifetime prevalence of allergic asthma ($p<0.001$) and trended to have a higher lifetime prevalence of allergic rhinitis ($p=0.06$) (table 1).

The final multivariate logistic regression model including the total revealed a significant association between ADHD and AE (odds ratio (OR) 1.54; 95% CI 1.24 to 1.93; $p<0.001$) that was independent of comorbid allergic rhinitis (OR 0.82 95% CI 0.63 to 1.06) or asthma (OR 1.23 95% CI 0.87 to 1.72) and all further investigated environmental factors. Interestingly, positive parental history of atopy was also independently related with being ever diagnosed as having ADHD (OR 1.41; 95% CI 1.18 to 1.69) (table 2).

The preplanned subgroup analysis confirmed that the association between ADHD and AE is modified by sleeping problems (interaction effect AE $\times$ sleep problems OR 2.02; 95% CI 1.03 to 3.97; $p=0.04$; table 3). Stratified analyses suggested a strong association between AE and ADHD in children with sleeping problems (OR 2.67; 95% CI 1.51 to 4.71; $p=0.001$; $n=1112$), but not in children without sleeping problems (OR 1.24; 95% CI 0.83 to 1.84; $p=0.30$; $n=5796$). No further significant interaction effects were found.

Table 1 Sample characteristics

Characteristic	Total ($n=13\,318$)			Atopic eczema				p Value	Attention-deficit/hyperactivity disorder				p Value
	n	%	Miss	Yes ($n=1952$)	%	No ($n=11\,366$)	%		Yes ($n=653$)	%	No ($n=12\,665$)	%	
Mean (SD) age (years)	9.9 (4.2)		0	9.9 (4.1)		9.9 (4.3)		0.74	11.8 (3.4)		9.8 (4.3)		<0.001
Female sex	6663	50.0	0	994	50.9	5669	49.9	0.39	133	20.4	6530	51.6	<0.001
Family income per month			595										
<€1000	896	7.0		110	5.9	786	7.2	<0.001	61	9.9	835	6.9	<0.001
€1000–1999	3560	28.0		466	24.9	3094	28.5		220	35.6	3340	27.6	
€2000–2999	4835	38.0		704	37.7	4131	38.1		205	33.2	4630	38.3	
≥€3000	3432	27.0		590	31.6	2842	26.2		132	21.4	3300	27.3	
Social status			56										
Low	3392	25.6		411	21.1	2981	26.4	<0.001	218	33.5	3174	25.2	<0.001
Intermediate	6324	47.7		914	46.9	5410	47.8		318	48.9	6006	47.6	
High	3546	26.7		623	32.0	2923	25.8		114	17.5	3432	27.2	
Rural/small town environment	6659	50.0	0	953	48.8	5706	50.2	0.26	342	52.4	6317	49.9	0.21
Living in East Germany	4479	33.6	0	684	35.0	3795	33.4	0.15	228	34.9	4251	33.6	0.48
Parental history of atopy (≥ 1 positive)	5717	43.1	61	1131	58.2	4586	40.5	<0.001	307	47.2	5410	42.9	0.030
No siblings	2149	16.3	133	332	17.2	1817	16.2	0.26	93	15.5	2056	16.4	0.20
Breastfeeding (ever)	10 184	77.6	185	1557	80.6	8627	77.0	0.001	448	70.2	9736	77.9	<0.001
Exposure to tobacco smoke at home	3992	30.3	120	501	25.9	3491	31.0	<0.001	273	42.0	3719	29.4	<0.001
Maternal smoking during pregnancy	2212	16.9	188	255	13.3	1957	17.5	<0.001	156	24.6	2056	16.5	<0.001
Postnatal health problems	3139	23.8	122	532	27.5	2607	23.2	<0.001	233	36.4	2906	23.1	<0.001
Allergic rhinitis (ever)	1550	11.6	0	452	23.2	1098	9.7	<0.001	91	13.9	1459	11.5	0.060
Allergic asthma (ever)	683	5.1	4	245	12.6	438	3.9	<0.001	53	8.1	630	5.0	<0.001

Table 2 Logistic regression on attention-deficit/hyperactivity disorder (n=12 167 study subjects without missing data)

Outcome variable (reference)	Bivariate (unadjusted) analysis		Multivariate (adjusted) analysis	
	OR (95% CI)	p Value	OR (95% CI)	p Value
Atopic eczema (absent vs present)	1.51 (1.24 to 1.84)	<0.001	1.54 (1.24 to 1.93)	<0.001
Age (years)	1.08 (1.06 to 1.11)	<0.001	1.09 (1.06 to 1.11)	<0.001
Sex (female)	4.16 (3.43 to 5.05)	<0.001	4.40 (3.57 to 5.42)	<0.001
Family income per month (<€1000)				
€1000–1999	0.90 (0.67 to 1.21)	0.489	0.92 (0.67 to 1.26)	0.602
€2000–2999	0.61 (0.45 to 0.81)	0.001	0.66 (0.47 to 0.93)	0.019
≥€3000	0.55 (0.40 to 0.75)	<0.001	0.69 (0.46 to 1.04)	0.075
Social status (low)				
Intermediate	0.77 (0.65 to 0.92)	0.004	0.93 (0.74 to 1.16)	0.508
High	0.48 (0.38 to 0.61)	<0.001	0.61 (0.43 to 0.87)	0.006
Maternal smoking during pregnancy (never)	1.65 (1.37 to 1.99)	<0.001	1.34 (1.07 to 1.68)	0.010
Exposure to tobacco smoke at home (absent)	1.72 (1.46 to 2.02)	<0.001	1.33 (1.10 to 1.62)	0.004
Perinatal health problems (absent)	1.90 (1.61 to 2.24)	<0.001	1.73 (1.45 to 2.07)	<0.001
Breastfeeding (never)	0.67 (0.56 to 0.80)	<0.001	0.86 (0.71 to 1.05)	0.145
Siblings (none vs any)	0.86 (0.69 to 1.08)	0.203	0.79 (0.62 to 1.01)	0.063
Allergic rhinitis (absent)	1.24 (0.99 to 1.56)	0.061	0.82 (0.63 to 1.06)	0.128
Allergic asthma (absent)	1.69 (1.26 to 2.26)	<0.001	1.23 (0.87 to 1.72)	0.237
Parental history of atopy (absent)	1.19 (1.02 to 1.39)	0.030	1.41 (1.18 to 1.69)	<0.001

DISCUSSION

Statement of principal findings

We replicated our recent finding of an association between ADHD and AE in a large population-based sample, adding substantial evidence to the notion that the observed relationship is true.¹⁵ Whereas the association was independent of various factors that are known to have impact either on the presence of AE and/or ADHD, we identified sleep impairment as the single relevant modifier of the finding. For children with sleeping problems and AE, the risk of being diagnosed as having ADHD was approximately 2.5-fold higher, whereas AE and ADHD were not significantly associated in children without sleep impairment.

Strengths and weaknesses of the study

Strengths of the study include the fact that it is based on a large population-based sample with a high recruitment rate suggesting high external validity and the extended availability on a large set of potential confounders, enabling us to eliminate various possible factors to underlie the observation. However, residual confounding cannot be completely ruled out. Based on the available data, we cannot determine the way in which sleep is disturbed.

Since the information with regard to previously met medical diagnoses is solely based on parental report, the validity and reliability of diagnoses cannot be determined and may limit the interpretation of the results. However, the prevalence rates found for ADHD and AE in the KIGGS database are commensurate to rates generally reported in the literature^{3–9} indicating that the survey accurately captured the disorders under investigation.

Since we applied a correlational study design, no definite causal interpretation can be deduced from the data.

Strengths and weaknesses in relation to other studies, discussing important differences in results

Early reports regarding the co-occurrence of allergic disorders and ADHD are conflicting. Although, in one study, higher rates of attentional problems and disruptive behaviour were found in children with atopic eczema compared with controls,²⁷ other samples of children with a diagnosis of ADHD did not display increased rates of allergic disorders.²⁸ Investigations with regard to an association of ADHD and asthma returned negative results and postulated an independent familial transmission of both disorders.^{29–30} In contrast, 30 clinically referred children and adolescents with ADHD displayed high

Table 3 Logistic regression on attention-deficit/hyperactivity disorder in a subsample with extended data availability (n=6484 study subjects) and stratified analysis by the presence of sleep problems

Outcome variable (reference)	Total		Stratified by sleep problems			
	OR (95% CI)	p Value	With sleep problems		Without sleep problems	
Atopic eczema (absent vs present)	1.22 (0.82 to 1.82)	0.33	2.67 (1.51 to 4.71)	<0.001	1.24 (0.83 to 1.84)	0.30
Sleep problems	2.07 (1.49 to 2.87)	<0.001	—	—	—	—
Interaction atopic eczema×sleep problems	2.02 (1.03 to 3.97)	0.04	—	—	—	—
Age (years)	1.34 (1.26 to 1.43)	<0.001	1.46 (1.29 to 1.66)	<0.001	1.30 (1.20 to 1.40)	<0.001
Sex (female)	3.80 (2.80 to 5.16)	<0.001	3.95 (2.23 to 7.00)	<0.001	3.72 (2.59 to 5.34)	<0.001
Social status (low)						
Intermediate	0.66 (0.50 to .90)	0.001	0.93 (0.52 to 1.64)	0.79	0.58 (0.41 to 0.82)	0.002
High	44 (0.30 to 0.65)	<0.001	51 (0.24 to 1.08)	0.08	41 (0.26 to 0.66)	<0.001
Exposure to tobacco smoke at home (absent)	1.82 (1.39 to 2.40)	<0.001	2.09 (1.24 to 3.53)	0.005	1.71 (1.24 to 2.38)	0.001
Perinatal health problems (absent)	1.89 (1.45 to 2.48)	<0.001	2.51 (1.52 to 4.15)	<0.001	1.68 (1.22 to 2.38)	0.002
Parental history of atopy (absent)	1.40 (1.07 to 1.82)	0.01	1.28 (0.77 to 2.12)	0.34	1.46 (1.07 to 2.00)	0.017

rates of clinically manifest allergic rhinitis, positive skin prick tests and atopic family history; however, no control group was investigated.⁵¹

The controlled trials by Egger *et al*^{32 33} suggested that at least a subsample of children with diagnosis of ADHD is allergic (or rather oversensitive) to certain food components resulting in increased motor activity and attentional deficits, and so dietary restrictions may have therapeutic benefit. A meta-analysis on 15 double-blind controlled trials on the behavioural effects of artificial food colourings in children with ADHD suggested a small effect on hyperactive behaviour and attention.³⁴ Recent findings of a controlled trial in healthy children support this notion;³⁵ however, the clinical relevance of the observation remains unclear,³⁶ and it is as yet unknown whether allergic mechanisms underlie the suggested responsiveness to food components.⁶ Although, in this regard, we had no information on alimentary habits, the strong association between AE and ADHD observed in our study can hardly be explained by the reportedly small effect exerted by the consumption of artificial food colourings.

In our previous report on the association of AE and ADHD, we suggested that the relative risk of being diagnosed as having ADHD increased with symptom severity of AE. Concomitant presence of sleep problems are suggestive as an indication of more severe AE; however, we were unable to test for this hypothesis due to the absence of a valid assessment of symptom severity in AE in our study.³⁷

Our approach confirmed and corroborated the recently reported association of AE and ADHD in an epidemiological sample. Thus, inconsistent results with regard to the presumptive connection between allergy and ADHD in general might have been confounded by the presence of AE, since neither sample sizes nor methods of previous studies allowed for AE to be sufficiently controlled. Furthermore, we found that sleep disturbance modified the effect, while ruling out various other possible influencing factors.

Meaning of the study: possible explanations and implications for clinicians and policymakers

ADHD is a multifactorial and heterogeneous disorder with a high familiarity and heritability.^{1 8 23} Assuming a complex aetiology, genetic factors account for approximately 80% of the phenotypic variance,²³ whereas some environmental factors, such as prenatal maternal smoking and low birth weight, have been identified as being associated with an increased risk of developing ADHD.^{16 17} AE, also, is clinically heterogeneous and aetiologically complex, and results from both genetic and environmental causes.⁹ On this background, our results might be interpreted in different ways.

1. Since AE most frequently occurs before ADHD symptoms develop to the full clinical picture *and* since AE is known to be a frequent cause for sleep disturbance in early childhood *and* since sleep disruption may result in disturbance of memory, poor concentration and disruptive behaviour,^{10 22 38 39} AE may be causing or exacerbating ADHD symptoms ("AE-induced ADHD").
2. ADHD symptoms cause substantial psychosocial impairment and are accompanied by a wide range of comorbid disorders such as depression, anxiety, as well as disturbance of sleep.^{5 19 20} Since psychosocial and emotional stress affects the severity of AE,²¹ ADHD might be an underlying cause or exacerbating factor for AE ("ADHD-induced AE").
3. ADHD, AE and sleep disturbance share common aetiological (genetic or environmental or both) factors. In this case, all three disorders/domains might be interpreted as manifesta-

tions of a syndromal entity based on the suggestions by Jensen *et al*⁴ ("syndrome").

4. All three previous statements are applicable, and ADHD and AE are associated in a complex gene–environment interaction mediated by sleeping disorders ("complex interaction").

Due to the cross-sectional design of our study, no definite causal conclusion can be drawn in this regard. However, considering the complexity of interaction of both AE and ADHD with already known aetiological factors, we believe the effect is most probably caused by complex interaction. Parental history of atopy was associated with ADHD in the primary regression analysis possibly indicating a common genetic background of AE and ADHD; however, testing for this assumption requires large genetic studies.

Our results suggest that there may be a definite subsample of children with severe health impairment due to multiple affection with psychiatric, dermatological and sleep-related problems being at high risk for the development of comorbid disorders. These children may require combined and integrated medical diagnostic, prevention and treatment strategies. This is highly relevant for public health and for health finances, since both AE and ADHD impose a high financial burden on social services.^{2 40}

Unanswered questions and future research

However, in order to deduct valid recommendations for clinical practise and public health services, further research on the causal relationships between AE, ADHD and concomitant sleep disturbance is required. Based on the available data, we cannot determine the specific nature of sleep disturbance. Despite our increasing understanding of the multidirectional impact of psychoneuroimmunology linking sleep, depression and stress with immune function, to date the presumptive aetiological mechanisms that may explain our finding remain speculative.⁴¹ In order to determine the developmental trajectories of AE, ADHD and SP, we propose the inclusion of child psychiatric and dermatological assessments in birth cohort studies.

What is already known on this subject

- Atopic eczema (AE) and attention-deficit/hyperactivity disorder (ADHD) are frequent and highly impairing childhood conditions imposing a substantial financial burden on social services.
- Recently, an association between AE and ADHD was reported.

What this study adds

- Whereas atopic comorbidity and various environmental factors did not account for the replicated association of AE and ADHD, the presence of sleep problems modified the relationship.
- Children with AE and concomitant sleep disturbance have a more than 2.5-fold increased risk of being diagnosed as having ADHD.
- The relationship between AE and ADHD modified by sleep problems might be of high relevance for public health and requires aetiological clarification to identify adequate preventive and therapeutic strategies.

Acknowledgements We thank the Robert-Koch-Institute, for making the data of the KIGGS study available, and all participants, for their time and information.

Funding The study was accomplished within the framework of the Clinical Research Program ADHD (KFO 125) funded by the German Research Association.

Competing interests None.

Contributors All authors planned the study. Statistical analysis was conducted by JS. All authors discussed and interpreted the results. MR drafted the manuscript and is guarantor. All authors contributed to the final manuscript.

Provenance and peer review Not commissioned; not externally peer reviewed.

REFERENCES

- Romanos M, Freitag C, Jacob C, *et al.* Genome-wide linkage analysis of ADHD using high-density SNP arrays: novel loci at 5q13.1 and 14q12. *Mol Psychiatry* 2008;**13**:522–30.
- Birnbaum HG, Kessler RC, Lowe SW, *et al.* Costs of attention deficit-hyperactivity disorder (ADHD) in the US: excess costs of persons with ADHD and their family members in 2000. *Curr Med Res Opin* 2005;**21**:195–206.
- Polanczyk G, de Lima MS, Horta BL, *et al.* The worldwide prevalence of ADHD: a systematic review and meta-regression analysis. *Am J Psychiatry* 2007;**164**:942–8.
- Jensen PS, Martin D, Cantwell DP. Comorbidity in ADHD: implications for research, practice, and DSM-V. *J Am Acad Child Adolesc Psychiatry* 1997;**36**:1065–79.
- Kadesjo B, Gillberg C. The comorbidity of ADHD in the general population of Swedish school-age children. *J Child Psychol Psychiatry* 2001;**42**:487–92.
- Pelsser L, Buitelaar J, Savelkoul H. ADHD as a (non) allergic hypersensitivity disorder: A hypothesis. *Pediatr Allergy Immunol* 2008;**20**:107–12.
- Johansson SG, Bieber T, Dahl R, *et al.* Revised nomenclature for allergy for global use: Report of the Nomenclature Review Committee of the World Allergy Organization, October 2003. *J Allergy Clin Immunol* 2004;**113**:832–6.
- Schmitt J, Romanos M. Lack of studies investigating the association of childhood eczema, sleeping problems, and attention-deficit/hyperactivity disorder. *Pediatr Allergy Immunol* 2009;**20**:299–300.
- Williams HC. Clinical practice. Atopic dermatitis. *N Engl J Med* 2005;**352**:2314–24.
- Smaldone A, Honig JC, Byrne MW. Sleepless in America: inadequate sleep and relationships to health and well-being of our nation's children. *Pediatrics* 2007;**119**:29–37.
- Carroll CL, Balkrishnan R, Feldman SR, *et al.* The burden of atopic dermatitis: impact on the patient, family, and society. *Pediatr Dermatol* 2005;**22**:192–9.
- Lewis-Jones S. Quality of life and childhood atopic dermatitis: the misery of living with childhood eczema. *Int J Clin Pract* 2006;**60**:984–92.
- Spergel JM, Paller AS. Atopic dermatitis and the atopic march. *J Allergy Clin Immunol* 2003;**112**:S118–27.
- Flohr C, Johansson SG, Wahlgren CF, *et al.* How atopic is atopic dermatitis? *J Allergy Clin Immunol* 2004;**114**:150–8.
- Schmitt J, Romanos M, Schmitt NM, *et al.* Atopic eczema and attention-deficit/hyperactivity disorder in a population-based sample of children and adolescents. *JAMA* 2009;**301**:724–6.
- Mick E, Biederman J, Faraone SV, *et al.* Case-control study of attention-deficit hyperactivity disorder and maternal smoking, alcohol use, and drug use during pregnancy. *J Am Acad Child Adolesc Psychiatry* 2002;**41**:378–85.
- Mick E, Biederman J, Prince J, *et al.* Impact of low birth weight on attention-deficit hyperactivity disorder. *J Dev Behav Pediatr* 2002;**23**:16–22.
- Kurth BM, Kamtsiuris P, Hölling H, *et al.* The challenge of comprehensively mapping children's health in a nation-wide health survey: design of the German KiGGS-Study. *BMC Public Health* 2008;**8**:196.
- Cortese S, Konofal E, Yateman N, *et al.* Sleep and alertness in children with attention-deficit/hyperactivity disorder: a systematic review of the literature. *Sleep* 2006;**29**:504–11.
- Sadeh A, Pergamin L, Bar-Haim Y. Sleep in children with attention-deficit hyperactivity disorder: a meta-analysis of polysomnographic studies. *Sleep Med Rev* 2006;**10**:381–98.
- Langan SM, Bourke JF, Silcocks P, *et al.* An exploratory prospective observational study of environmental factors exacerbating atopic eczema in children. *Br J Dermatol* 2006;**154**:979–80.
- Paavonen EJ, Räikkönen K, Lahti J, *et al.* Short sleep duration and behavioral symptoms of attention-deficit/hyperactivity disorder in healthy 7- to 8-year-old children. *Pediatrics* 2009;**123**:e857–64.
- Biederman J, Faraone SV. Attention-deficit hyperactivity disorder. *Lancet* 2005;**366**:237–48.
- Swanson JM, Sergeant JA, Taylor E, *et al.* Attention-deficit hyperactivity disorder and hyperkinetic disorder. *Lancet* 1998;**351**:429–33.
- Lange M, Kamtsiuris P, Lange C, *et al.* Sociodemographic characteristics in the German Health Interview and Examination Survey for Children and Adolescents (KiGGS)—operationalisation and public health significance, taking as an example the assessment of general state of health. *Bundesgesundheitsblatt Gesundheitsforschung Gesundheitsschutz* 2007;**50**:578–89.
- Rosner B. *Fundamentals of biostatistics*. 5th edn. Pacific Grove: Duxbury, 2000.
- Roth N, Beyreiss J, Schlenzka K, *et al.* Coincidence of attention deficit disorder and atopic disorders in children: empirical findings and hypothetical background. *J Abnorm Child Psychol* 1991;**19**:1–13.
- Mitchell EA, Aman MG, Turbott SH, *et al.* Clinical characteristics and serum essential fatty acid levels in hyperactive children. *Clin Pediatr (Phila)* 1987;**26**:406–11.
- Biederman J, Milberger S, Faraone SV, *et al.* Associations between childhood asthma and ADHD: issues of psychiatric comorbidity and familiarity. *J Am Acad Child Adolesc Psychiatry* 1994;**33**:842–8.
- Hammeress P, Monuteaux MC, Faraone SV, *et al.* Reexamining the familial association between asthma and ADHD in girls. *J Atten Disord* 2005;**8**:136–43.
- Brawley A, Silverman B, Kearney S, *et al.* Allergic rhinitis in children with attention-deficit/hyperactivity disorder. *Ann Allergy Asthma Immunol* 2004;**92**:663–7.
- Egger J, Carter CM, Graham PJ, *et al.* Controlled trial of oligoantigenic treatment in the hyperkinetic syndrome. *Lancet* 1985;**1**:540–5.
- Egger J, Stolla A, McEwen LM. Controlled trial of hyposensitisation in children with food-induced hyperkinetic syndrome. *Lancet* 1992;**339**:1150–3.
- Schab DW, Trinh NH. Do artificial food colors promote hyperactivity in children with hyperactive syndromes? A meta-analysis of double-blind placebo-controlled trials. *J Dev Behav Pediatr* 2004;**25**:423–34.
- McCann D, Barrett A, Cooper A, *et al.* Food additives and hyperactive behaviour in 3-year-old and 8/9-year-old children in the community: a randomised, double-blinded, placebo-controlled trial. *Lancet* 2007;**370**:1560–7.
- European Food Safety Authority. Scientific Opinion of the Panel on food additives, flavourings, processing Aids and Food Contact materials (AFC) on a request from the commission on the results of the study by McCann, *et al.* (2007) on the effect of some colours and sodium benzoate on children's behaviour. *EFSA J* 2008;**660**:1–54.
- Schmitt J, Langan S, Williams HC, *et al.* What are the best outcome measurements for atopic eczema? A systematic review. *J Allergy Clin Immunol* 2007;**120**:1389–98.
- Boonstra TW, Stins JF, Daffertshofer A, *et al.* Effects of sleep deprivation on neural functioning: an integrative review. *Cell Mol Life Sci* 2007;**64**:934–46.
- Stickgold R. Sleep-dependent memory consolidation. *Nature* 2005;**437**:1272–8.
- Jenner N, Campbell J, Marks R. Morbidity and cost of atopic eczema in Australia. *Australas J Dermatol* 2004;**45**:16–22.
- Irwin MR. Human psychoneuroimmunology: 20 years of discovery. *Brain Behav Immun* 2008;**22**:129–39.